

Aviv Segall

Software Engineer and Computer
Graphics Researcher

* 3 May 1991
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Education

- 2022–2026 **Ph.D. in Computer Graphics**, *ETH Zurich*, Zurich
Dissertation: *Inverse Design of Freeform Surfaces via Reconfigurable Tessellations*
Advisor: Prof. Olga Sorkine-Hornung
- 2014–2016 **M.S. in Computer Graphics**, *Technion - Israel Institute of Technology*, Haifa, 95
Thesis: *2D Simulation and Mapping using the Cauchy-Green Complex Barycentric Coordinates*
Advisor: Prof. Mirela Ben-Chen
- 2011–2014 **B.S. in Computer Science**, *Technion - Israel Institute of Technology*, Haifa, 97

Academic and Research Experience

- 2022–current **Doctoral Researcher**, *ETH Zurich*, Zurich
Researched the connection between origami tessellations, hinged kirigami structures and smocking (a fabric manipulation technique), leading to the development of computational tools for designing deployable patterns that can approximate a target 3D shape upon deployment. Potential applications include mechanical metamaterial design, shock-absorbing packaging, soft robotics and architecture.
- 2018–2022 **Full-Stack Engineer**, *Google*, Zurich
- Worked with several teams across the globe to improve the quality of the user's location inference by implementing a strong signal for a correct inference.
 - Conducted several experiments to find the best balance between the quality and the amount of collected data for improving the location inference.
 - Developed an Android framework for communicating between the native app and a web page running under a WebView.
 - Worked on the integration of the user's location history in Android Google Photos.
- 2017–2018 **Backend Engineer**, *Waze*, Tel Aviv
- Worked on the backend side of the Waze Carpool app, developing algorithms for matching between potential drivers and riders.
 - Took part in an on-call rotation, monitoring the service's health, responding to and fixing bugs.
 - Collaborated with the frontend team to define APIs and develop a feature from design to deployment in production.
 - Worked on improving the quality of the voice commands in Waze, analyzing failure cases and suggesting improvements to the scoring function to provide better recognition.
- 2013–2016 **Master Student and Teaching Assistant**, *Technion - Israel Institute of Technology*, Haifa
- Master student in computer graphics, focusing on applications of the Cauchy-Green coordinates (discretization of the Cauchy Integral Formula) for simulating Hele-Shaw flows and finding conformal maps between two 2D domains.
 - Teaching assistant in the computer science faculty (taught the courses "Introduction to Computer Science" and "Computer Organization and Programming")

- 2013–2013 **Software Engineer Intern**, *Intel*, Haifa
Worked on the integration of a SAT solver for scheduling processor tests on several machines.
- 2011–2013 **Student Software Engineer**, *Intel*, Haifa
Developed instrumentation tools for a processor simulator.

Publications

- [1] Annika Oehri, **Aviv Segall**, Amir Vaxman, and Olga Sorkine-Hornung. Fabricating 3d shapes by shrinking-thread embroidery. In *SIGGRAPH Asia Conference Papers '26*, SIGGRAPH Asia Conference Papers '26, New York, NY, USA, 2026. Association for Computing Machinery.
- [2] **Aviv Segall**, Jing Ren, and Olga Sorkine-Hornung. Uniformly deployable kirigami on arbitrary planar graphs. *ACM Transactions on Graphics (TOG)*, 45(4):1–17, 2026.
- [3] Ruben Wiersma, Alexandre Binniger, Peizhuo Li, Tanguy Magne, Annika Oehri, **Aviv Segall**, Danielle Luterbacher, Marcel Padilla, Jing Ren, and Olga Sorkine-Hornung. Gradient descent in the alps: Abstracted low-poly stylization and fabrication. In *Proceedings of the Special Interest Group on Computer Graphics and Interactive Techniques Conference Conference Papers*, pages 1–11, 2026.
- [4] **Aviv Segall**, Jing Ren, Marcel Padilla, and Olga Sorkine-Hornung. Reconfigurable hinged kirigami tessellations. In *Proceedings of the SIGGRAPH Asia 2025 Conference Papers*, pages 1–11, 2025.
- [5] **Aviv Segall**, Jing Ren, Martin Schwarz, and Olga Sorkine-Hornung. Computational modeling of gothic microarchitecture. In *Proceedings of the Special Interest Group on Computer Graphics and Interactive Techniques Conference Conference Papers*, pages 1–11, 2025.
- [6] Annika Oehri, **Aviv Segall**, Jing Ren, and Olga Sorkine-Hornung. Chebyshev parameterization for woven fabric modeling. *ACM Transactions on Graphics (TOG)*, 43(6):1–15, 2024.
- [7] **Aviv Segall**, Jing Ren, Amir Vaxman, and Olga Sorkine-Hornung. Fabric tessellation: Realizing freeform surfaces by smocking. *ACM Transactions on Graphics (TOG)*, 43(4):1–20, 2024.
- [8] Jing Ren, **Aviv Segall**, and Olga Sorkine-Hornung. Digital 3d smocking design. *ACM Transactions on Graphics (TOG)*, 43(2):14:1–14:17, 2024. Presented at SIGGRAPH ASIA 2023.
- [9] **Aviv Segall** and Mirela Ben-Chen. Iterative closest conformal maps between planar domains. In *Computer Graphics Forum*, volume 35, pages 33–40. Wiley Online Library, 2016.
- [10] **Aviv Segall**, Orestis Vantzos, and Mirela Ben-Chen. Hele-shaw flow simulation with interactive control using complex barycentric coordinates. In *Symposium on Computer Animation*, volume 85, page 95, 2016.
- [11] **Aviv Segall**, Jonathan Mizrahi, Yong Joon Kim, and Gershon Elber. Line accessibility of free form surfaces. *Graphical models*, 76(5):301–311, 2014.

Languages

English Fluent
Hebrew Native

Computer skills

Programming languages C/C++, Java, Python, JavaScript, TypeScript, MATLAB
Development Environment Linux (Ubuntu/Arch), Windows (WSL2), CMake, Git, GDB

References

Prof. Dr. Olga Sorkine-Hornung, ETH Zurich (advisor), sorkine@inf.ethz.ch
Prof. Mirela Ben-Chen, Technion (advisor), mirela@cs.technion.ac.il
Prof. Amir Vaxman, The University of Edinburgh (advisor), avaxman@inf.ed.ac.uk